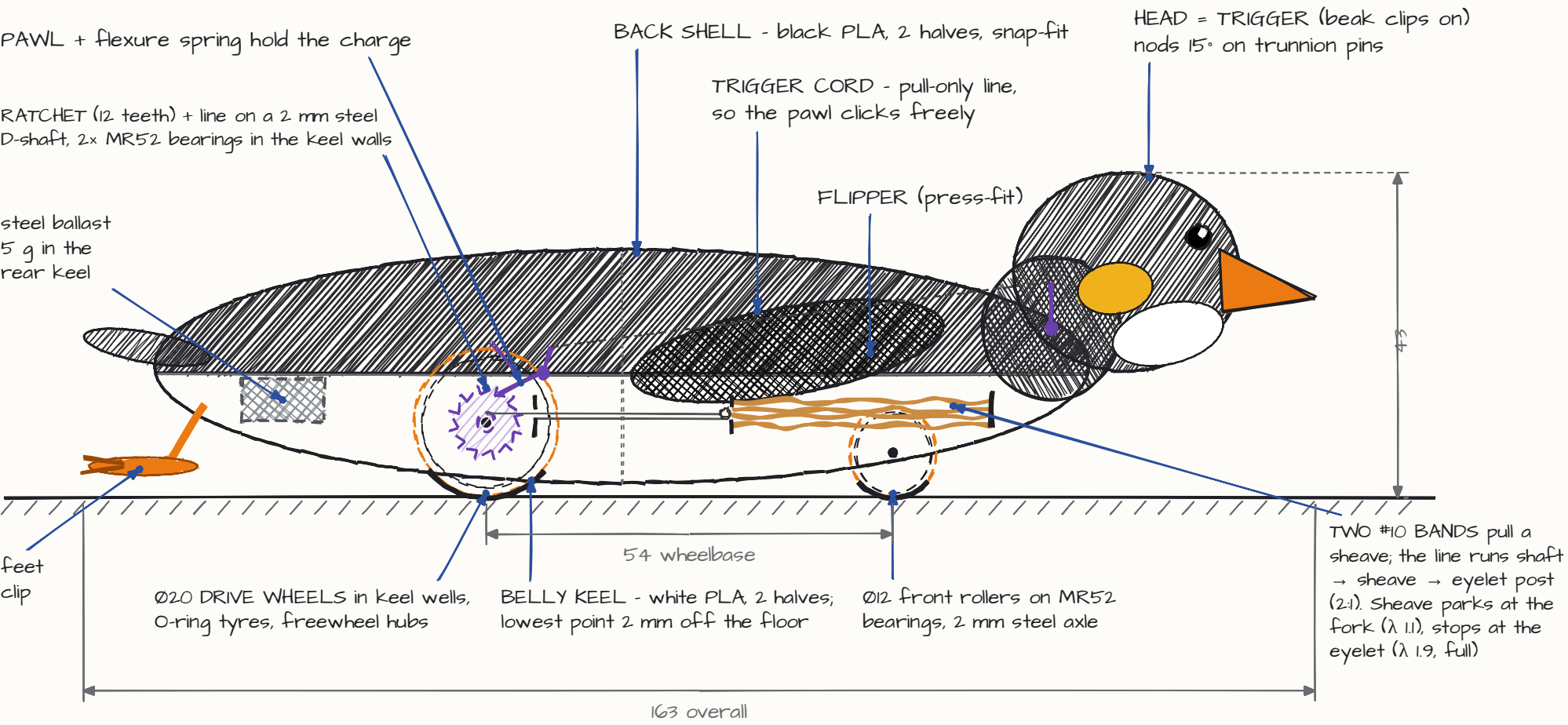
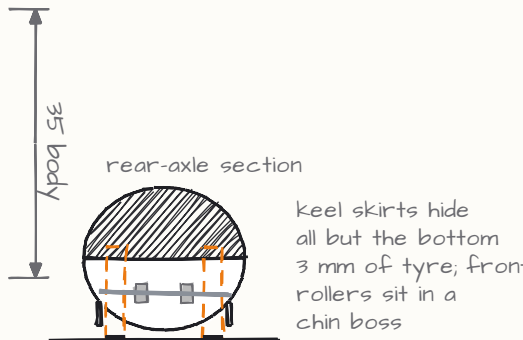
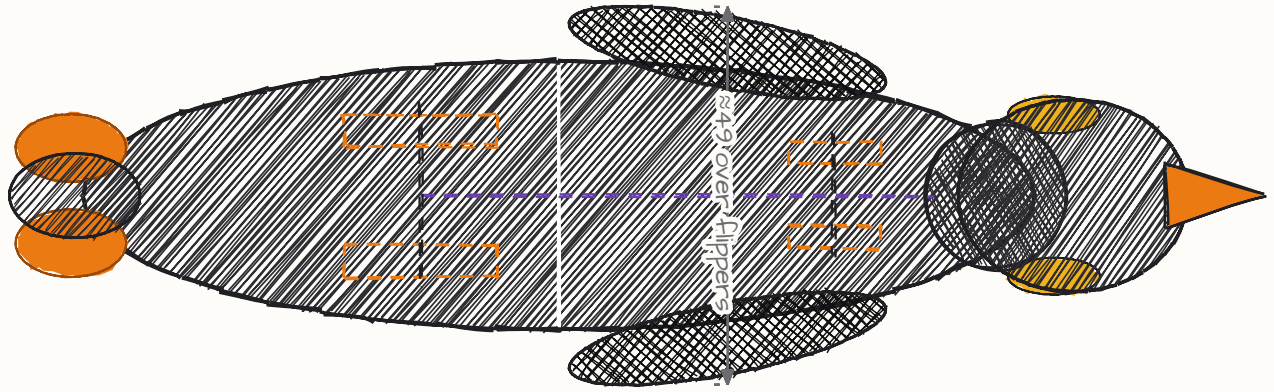


BODY - SIDE VIEW (scale 1:2.7 : 1 - dashed = hidden parts inside the body - purple = mechanism)

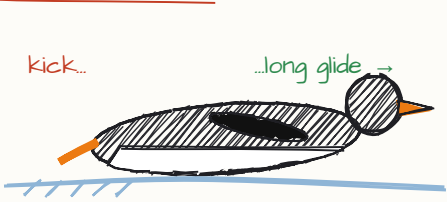


TOP VIEW

scale 1 : 1 (side 1:2.7 : 1)



BIOMIMICRY



Tobogganing penguins (emperor, Adélie) give a few kicks, then glide far on their bellies. Copied: a 22 cm power stroke (the band), then a long freewheel belly-glide on a keel; low, long body.

HOW IT PLAYS

- Click the head UP, press on its back and pull back ~43 cm
 - Let go - the ratchet holds it.
 - Tap the beak tip - it toboggans ~ 7 m (less if you pull less).
 - Lift the head to re-arm
- vs Concept 2: lying body; bands in TENSION + pawl; head-nod trigger.

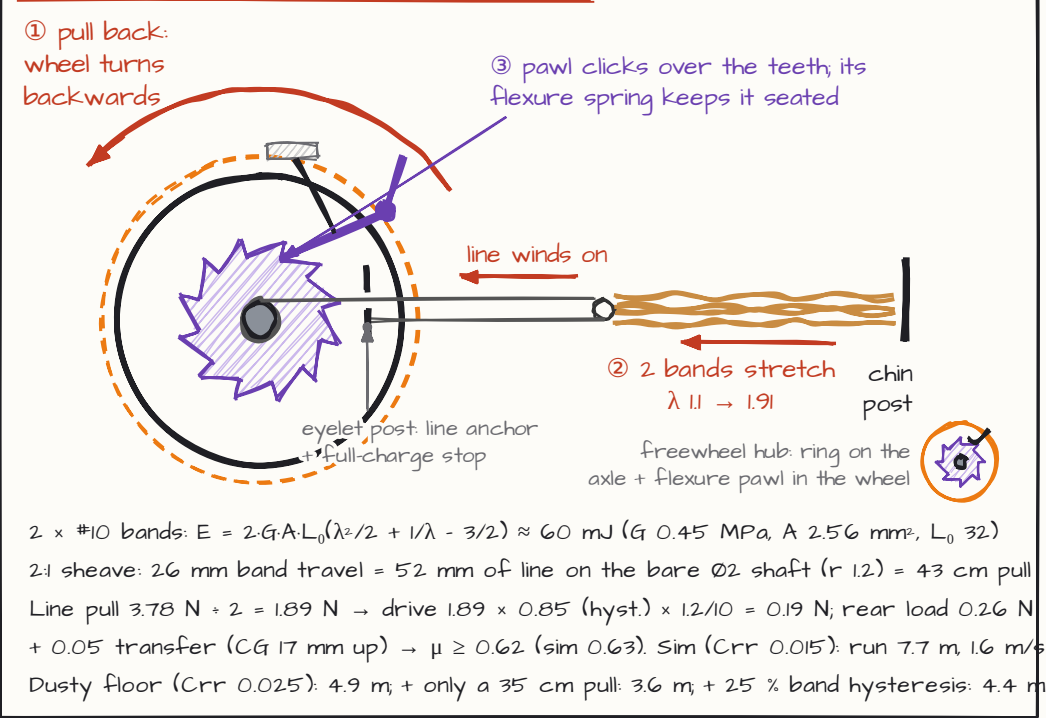
NUMBERS

- Mass ≈ 39 g, 78 % printed PLA
- CG 17 mm ahead of rear axle, 17 mm up (68 % on drive wheels)
- Bands store 60 mJ; sim run 7.7 m (dusty floor 4.9 m)
- Top speed ≈ 1.6 m/s
- No wheelspin above $\mu 0.63$ (nitrile O-ring ≈ 0.9 ; test it)
- Bought (17): 2 D-shafts, 6 MR52, 4 O-rings, 2 #10 bands, line, sheave pin, 5 g ballast. Printed: 18 parts

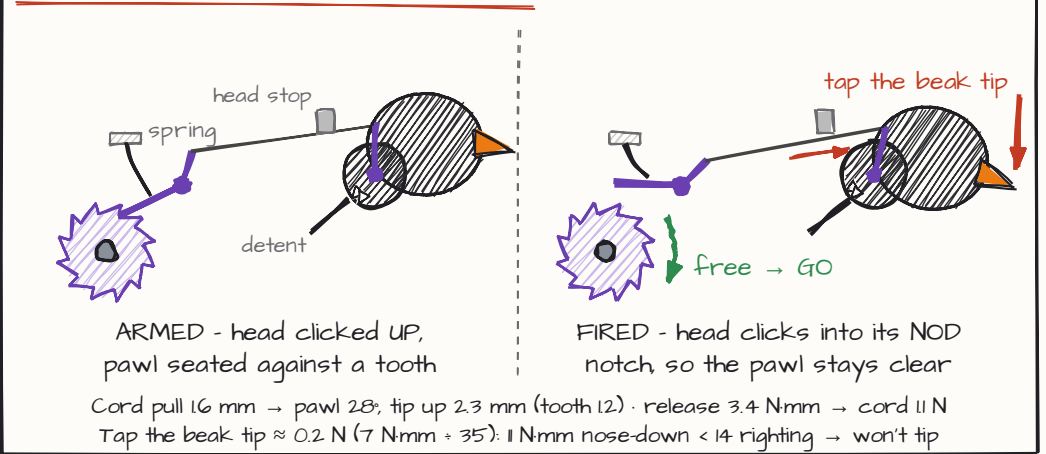
RULES CHECK

- ≥ 75 % printed PLA (78 %)
- Parts fit the egg (box C); built toy is bigger than the egg
- Stores energy and waits until the head is tapped
- Trigger is not a removable pin - nothing comes off
- No energy stored when apart (band relaxed)
- Snap / press fits - no glue, no tools needed
- No bought toy parts or downloaded CAD

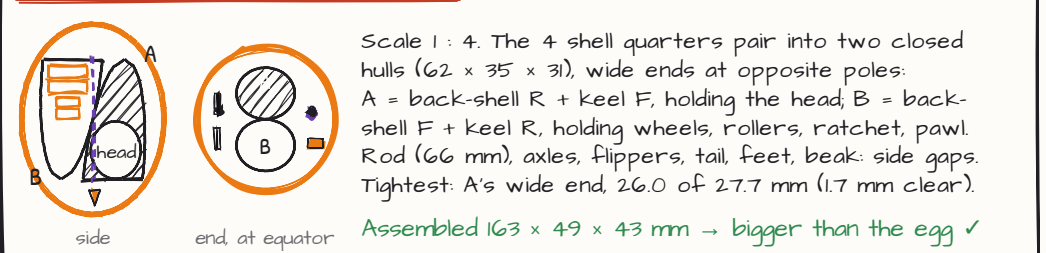
A ENERGY - PULL-BACK CHARGE



B TRIGGER - TAP THE HEAD



C PACKS INTO THE EGG



ENGG*2100 F26 - Design & Build - Concept Sketch

Concept 1 - sheet 1 of 1

Concept 1: TUX TOBOGGAN

Theme: Biomimicry - tobogganing penguin

Toy type: Running toy (≥ 2.5 m)

Scale / units: mm - side view 1:2.7 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____